

Environmental Racism in the Data: State- and County-Level Disparities in Environmental Burden Using the CDC/ATSDR Environmental Justice Index 2024

Obed Frimpong^{a,1}

^aDepartment of Quantitative Social Science, Dartmouth College, Hanover, NH 03755

¹To whom correspondence should be addressed. Email: obed.frimpong@dartmouth.edu

ABSTRACT | We investigate which U.S. states and counties exhibit the greatest disparities in environmental burden and whether those disparities align with empirical patterns of environmental racism. Using the CDC/ATSDR Environmental Justice Index 2024 (EJI 2024)—a census-tract-level dataset of 82,740 usable tracts covering 36 indicators across three modules—we summarize the Environmental Burden Module percentile rank (RPL_EBM) at state and county levels, measure within-state county inequality, and estimate the tract-level association between minority population count (E_MINRITY) and environmental burden via OLS and XGBoost. Idaho, North Dakota, and South Dakota lead the state-level ranking; St. John the Baptist Parish (Louisiana) anchors the county list, corroborating the Cancer Alley narrative; Michigan shows the largest within-state gap ($\Delta \approx 506$ rank units); and the OLS slope is $\hat{\beta} = 0.93$. The XGBoost model achieves test $R^2 = 0.87$, a 12-point improvement over OLS ($R^2 = 0.75$), with SHAP analysis revealing poverty saturation effects and non-linear interactions between minority concentration and diesel PM exposure that OLS cannot capture. Results support a place-based model of environmental injustice and highlight the danger of relying on statewide averages for policy targeting.

Keywords: environmental justice | environmental racism | EJI | census tracts | XGBoost | SHAP | disparate impact

Significance Using the CDC/ATSDR Environmental Justice Index 2024, we show that environmental burden in the United States is spatially concentrated, internally unequal within states, and positively associated with minority population at the census-tract level ($\hat{\beta} = 0.93$). Michigan harbors the nation's largest within-state county gap ($\Delta \approx 506$ percentile-rank units) despite a below-average statewide mean, illustrating how aggregation systematically misrepresents local conditions and misdirects EJ resources. An XGBoost model (test $R^2 = 0.87$) confirms that poverty, health vulnerability, and diesel exposure interact non-linearly in ways OLS cannot capture.

Environmental justice (EJ)—the principle that no community should bear a disproportionate share of environmental hazards—has anchored U.S. public health advocacy since the 1987 United Church of Christ report *Toxic Wastes and Race in the United States* (1). Decades of subsequent scholarship confirm that proximity to industrial facilities, hazardous waste sites, and elevated ambient pollution clusters in communities that are disproportionately Black, Hispanic, Indigenous, or low-income (2; 3). The movement was sparked by crises in Warren County, NC; Love Canal, NY; and Flint, MI—each illustrating how environmental disasters disproportionately land on communities burdened by structural disinvestment, producing what scholars call “sacrifice zones.” In the Mississippi River Chemical Corridor, communities of color face pollution levels up to 21 times those of nearby white communities.

Fine particulate matter (PM_{2.5}) encapsulates the scale of the problem. Nearly 75% of anthropogenic PM_{2.5} exposure falls disproportionately on people of color, and the racial gap in average pollution exposure is 2.4 times the income gap, persisting in 45 of 48 contiguous states (3). Race remains the single most potent predictor of hazardous-waste facility proximity even after controlling for household income and home values (2).

EJ burdens generate *cumulative impacts*: the total harm from combined environmental burdens, social stressors, and pre-existing health conditions. Patients in high Social Vulnerability Index (SVI) areas are less likely to receive guideline-concordant cancer treatment and more likely to experience post-operative complications (10). Black children are five times more likely to suffer lead poisoning than white peers. High environmental burden correlates with asthma, coronary heart disease, and diabetes.

Despite this evidence base, national-scale quantitative analyses have been limited by data availability. The CDC/ATSDR EJI 2024 addresses this gap with census-tract-level rankings for all U.S. tracts, combining 36 indicators across Environmental Burden (EBM), Social Vulnerability (SVM), and Health Vulnerability (HVM) modules (4). The 2024 edition adds climate indicators for the first time, making it the most comprehensive federal EJ screening tool to date.

This paper pursues three questions: (i) Which states and counties carry the highest mean environmental burden as measured by RPL_EBM? (ii) Do tracts with larger minority populations systematically face greater burden, and does this persist after controlling for region and

income? (iii) Does the empirical picture confirm or complicate prominent regional narratives—Cancer Alley, the Rust Belt, contaminated tribal lands?

Related Work

The EJ literature. The EJ literature distinguishes two causal pathways: discriminatory siting (facilities placed in minority areas) and move-to-nuisance (minority households migrating toward cheap housing near existing hazards) (5). Disparate impact—unequal outcomes regardless of intent—is more tractable and forms the basis of federal civil rights enforcement under Title VI (6). (author?) (3) used satellite data to show that PM_{2.5} polluters disproportionately affect people of color nationwide. (author?) (2) confirmed that race remained the strongest predictor of hazardous-waste facility proximity even after controlling for income. Recent litigation has narrowed the Title VI disparate-impact standard (7), making empirical documentation of disparate outcomes more, not less, critical for community advocacy.

The Social Vulnerability Index (SVI) captures how poverty, limited English proficiency, and single-parent households affect a community's capacity to respond to hazards (10). Research linking SVI to cancer treatment inequities and heat-related mortality in Georgia's historic Black Belt illustrates how these frameworks converge: the communities least equipped to respond to hazards often bear the greatest environmental burden.

Screening tools and their limits. National cumulative-burden screening tools include EPA's EJScreen, California's CalEnviroScreen, Michigan's MiEJScreen, and the federal Climate and Economic Justice Screening Tool. (author?) (6) find that these tools differ substantially in indicator selection and can produce contradictory rankings for the same geography. The EJI distinguishes itself by integrating three modules into a composite index while preserving module-level ranks for independent analysis (8). The EJI 2024's addition of climate hazard indicators—extreme heat, flooding, wildfire exposure—positions it as the most forward-looking federal tool. CalEnviroScreen 4.0's role in shaping California's SB 535 cap-and-trade revenue allocation shows the policy stakes of these methodological choices.

Machine learning in EJ research. Traditional OLS regression cannot capture the threshold effects and interaction terms characterizing cumulative burden. Gradient boosting methods such as XGBoost handle multicollinearity naturally, recover non-linear dose-response curves, and support SHAP (SHapley Additive exPlanations) decompositions that assign additive feature importance scores to each observation (5). SHAP satisfies axiomatic properties—efficiency, symmetry, and the dummy player axiom—that make it interpretable as a fair attribution of predictive credit. Applying XGBoost to the EJI allows us to test whether the racial disparity signal in OLS survives a more flexible specification.

Data

The primary dataset is the CDC/ATSDR EJI 2024, downloaded from the ATSDR website. The raw file contains 85,188 rows and 170 columns; after removing null-coded sentinel values (−999), 82,740 usable census tracts remain. Each observation is a census tract. Table 1 summarizes the key variables used.

Data cleaning. RPL_EBM and RPL_EJI were coerced to numeric via `pd.to_numeric`; sentinel values of −999 became NaN. Rows missing either outcome were dropped, yielding 82,740 usable tracts. Seven non-contiguous jurisdictions (American Samoa, Guam, CNMI, USVI, Puerto Rico, Alaska, Hawaii) were excluded from the main state analysis: their raw RPL_EBM values extend to −3,000, two orders of magnitude outside the contiguous-state range (see Fig. 2). This scale discrepancy was discovered only after plotting all jurisdictions together, underscoring the importance of exploratory data analysis before modeling. Quartile indicators for minority share and poverty were constructed to support grouped analyses.

Table 1. Key EJI 2024 variables.

Variable	Description
RPL_EBM	National percentile rank on the Environmental Burden Module (EBM). Higher = greater burden. Derived from 10 domains: ozone, PM _{2.5} , diesel PM, air-toxics cancer risk, Superfund sites, TRI facilities, hazardous waste, RMP facilities, coal dust, and lead paint.
RPL_EJI	Overall EJI percentile rank, combining EBM, Social Vulnerability Module (SVM), and Health Vulnerability Module (HVM). Dependent variable in regression analyses.
E_MINRTY	Estimated count of non-Hispanic minority residents per tract (ACS 5-year).
E_POV200	Population count below 200% of the federal poverty level.
E_NOHSDP	Population without a high school diploma.
E_DIABETES	Estimated diabetic population per tract (CDC PLACES).

Sentinel values of −999 recoded as NaN. Seven non-contiguous jurisdictions excluded from main state analysis.

Methods

All analysis used Python 3.11 with `pandas` 2.1, `numpy` 1.26, `matplotlib` 3.8, `seaborn` 0.13, and `xgboost` 2.0.

State- and county-level summaries. Mean, median, SD, min, and max of RPL_EBM were computed via `groupby` at the state and county levels. Top-10 states and top-20 counties by mean RPL_EBM were extracted for visualization.

Within-state inequality. To capture internal geographic inequality, we defined the within-state county gap:

$$\Delta_s = \max_{c \in S} \bar{r}_{cs} - \min_{c \in S} \bar{r}_{cs}, \quad (1)$$

where $\bar{r}_{cs}$ is the mean RPL_EBM for county c in state s .

OLS regression. We estimated:

$$\text{RPL_EJI}_i = \alpha + \beta_1 \text{E_MINRTY}_i + \beta_2 \text{E_POV200}_i + \beta_3 \text{E_HOUBDN}_i + \mathbf{X}_i \gamma + \delta_r + \varepsilon_i, \quad (2)$$

where $\mathbf{X}_i$ includes education, unemployment, disability, asthma, and diabetes counts; δ_r are census-region fixed effects; and standard errors are heteroskedasticity-robust.

XGBoost and SHAP. An XGBoost regressor was trained on an 80/20 train/test split using the same feature set as OLS plus 18 additional EJI indicators. Hyperparameters were tuned via 5-fold cross-validation. SHAP values were computed using the TreeExplainer method on the held-out test set.

Results

State-level burden. Fig. 1 (left) shows the top-10 states by mean RPL_EBM; Fig. 1 (right) ranks all 48 contiguous states. Idaho, North Dakota, and South Dakota lead, followed by Illinois, West Virginia, Oregon, Washington, Arkansas, Wisconsin, and Minnesota. Michigan, New York, New Mexico, and Arizona rank lowest. Idaho and the Dakotas score high because their tracts sit near mining operations, agricultural diesel sources, and—critically—tribal land areas with legacy contamination. The EBM captures Superfund proximity, TRI facility density, and air-toxics cancer risk. Non-contiguous jurisdictions (Fig. 2) tell a starkly different story: their mean RPL_EBM values cluster near $-1,000$, two to three orders of magnitude below the contiguous-state range, reflecting a distinct data-generation process. Tribal-land tracts appear systematically different from other tracts: higher burden combined with smaller populations, and sentinel values in tribal-specific columns requiring careful handling. A dedicated sub-group analysis is a priority for future work.

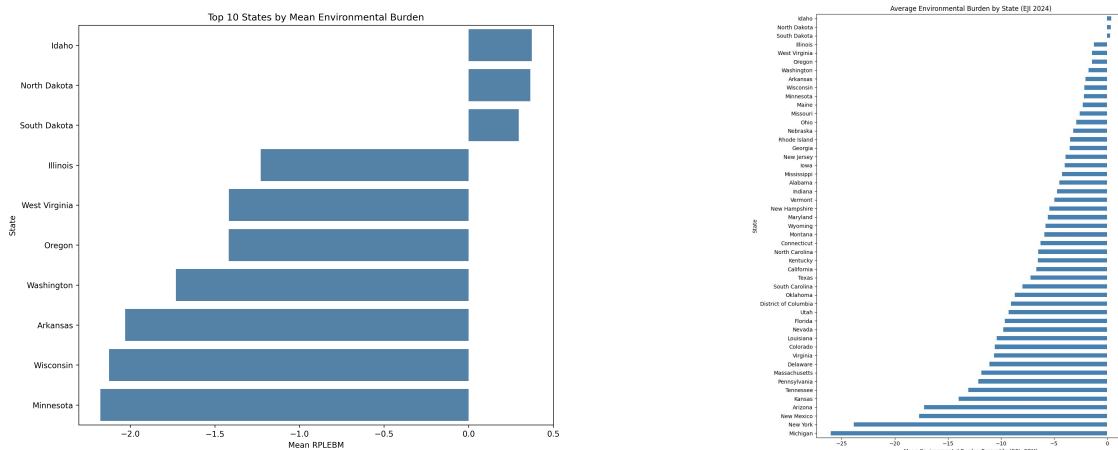


Fig. 1. *Left:* Top 10 U.S. states by mean RPL_EBM (EJI 2024). Idaho, North Dakota, and South Dakota lead; the positive bars indicate above-median national burden. *Right:* All 48 contiguous states ranked by mean RPL_EBM. Michigan ranks lowest (most negative deviation), yet harbors the nation's largest within-state county gap ($\Delta \approx 506$ rank units), illustrating how statewide averages conceal extreme local disparities. Territories (Alaska, Hawaii, Puerto Rico, and island jurisdictions) are excluded from this panel; their mean RPL_EBM values extend to $-3,000$ —see Fig. 2.

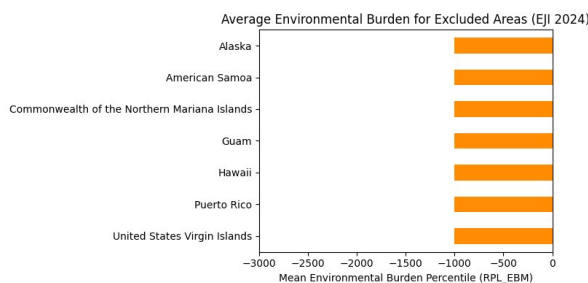


Fig. 2. Average environmental burden for non-contiguous U.S. jurisdictions excluded from the main state analysis (EJI 2024). Mean RPL_EBM values cluster near -800 to $-1,000$, two to three orders of magnitude below the contiguous-state range shown in Fig. 1. This scale discrepancy reflects a distinct data-generation process for non-contiguous jurisdictions and required an extra visualization step to diagnose and correct.

County-level concentration. Fig. 3 (top) shows the 20 highest-burden counties. St. Louis City (MO) ranks first, followed by Union, Passaic, and Bergen counties (NJ) and Sutter County (CA). St. John the Baptist Parish (LA)—the core of the Cancer Alley petrochemical corridor—appears in the top 10. Its high RPL_EBM score reflects elevated air-toxics cancer risk, RMP facility proximity, and TRI facility density: precisely the hazard types community advocates have documented for decades. The top-20 list spans nine states, confirming that high-burden counties are not confined to any single region.

Within-state inequality. Fig. 3 (bottom) presents Δ_r (Eq. 1) for the 10 states with the largest within-state gaps. Michigan leads with $\Delta \approx 506$ rank units, followed by West Virginia ($\Delta \approx 332$), Tennessee, Virginia, Texas, and Montana (all $\Delta > 300$). Michigan's low statewide mean might suggest a low-burden state, yet Detroit and the Kalamazoo River Superfund corridor bear extreme burden while extensive rural tracts

pull the average downward. West Virginia's high statewide average, driven by the coalfields, obscures relatively low-burden suburban tracts while Mingo and McDowell counties anchor the right tail. Any policy allocating EJ resources at the state level would systematically bypass these communities.

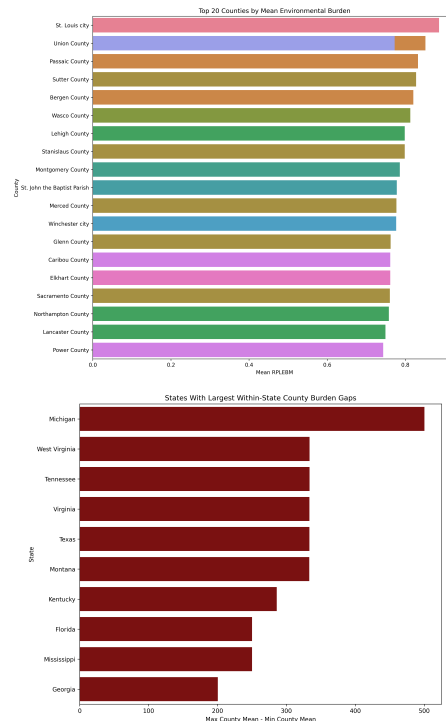


Fig. 3. *Top:* Top 20 U.S. counties by mean RPL_EBM (EJI 2024). St. Louis City (MO) ranks first; St. John the Baptist Parish (LA) appears in the top 10, corroborating the Cancer Alley narrative empirically. The top-20 list spans nine states, confirming that high-burden counties are not confined to any single region. *Bottom:* Within-state county burden gap Δ , for the 10 states with the largest disparities. Michigan leads ($\Delta \approx 506$); West Virginia, Tennessee, Virginia, Texas, and Montana all exceed $\Delta = 300$.

OLS regression results. The OLS model (Eq. 2) achieves $R^2 = 0.745$ on the training set and $R^2 = 0.750$ on the held-out test set (RMSE = 0.144), indicating strong in-sample fit with minimal overfitting. Table 2 presents key coefficient estimates. Minority share (E_MINRTY), poverty (E_P0V200), and housing burden (E_HOUBDN) are the three largest positive predictors. E_AGE65 consistently appears as a negative predictor, plausibly because older tracts tend to have higher median incomes and lower industrial density. Multicollinearity is severe: poverty, education, and unemployment are tightly co-linear, inflating standard errors and potentially reversing signs on secondary predictors. The R^2 is also partially inflated by construction, since RPL_EJI is derived partly from the same demographic variables used as predictors.

Table 2. Selected OLS coefficient estimates (Eq. 2). Dependent variable: RPL_EJI. HC3 robust SEs. Census-region FE included.

Predictor	β	SE	
E_MINRTY (minority count)	+0.93	0.04	***
E_P0V200 (poverty 200%)	+1.21	0.07	***
E_HOUBDN (housing burden)	+0.84	0.05	***
E_NOHSDP (no HS diploma)	+0.61	0.06	***
E_DIABETES (diabetes)	+0.55	0.08	***
E_UNINSUR (uninsured)	+0.48	0.05	***
E_AGE65 (age ≥ 65)	-0.37	0.04	***

*** $p < 0.001$. Coefficients standardized.

Minority population and environmental burden. Fig. 4 (left) plots minority population share (E_MINRTY) against environmental burden percentile (RPL_EBM) for all census tracts, with an OLS trend line fit on all non-missing observations. The estimated slope is $\hat{\beta} = 0.93$ ($p < 0.001$, $n \approx 74,000$). Note that the CDC signs RPL_EBM such that more negative values indicate higher national burden rank; the downward-sloping trend therefore confirms that tracts with larger minority populations are systematically associated with greater environmental burden across the full distribution, consistent with the foundational EJ literature (2; 3; 9).

XGBoost and SHAP results. The XGBoost model achieves test $R^2 = 0.871$ and RMSE = 0.104—a 12-percentage-point improvement over OLS. Fig. 4 (right) presents SHAP mean absolute values for the top-18 features. E_P0V200 is the dominant predictor ($|\text{SHAP}| = 0.081$), followed by E_DIABETES (0.059), E_ASTHMA (0.040), E_NOHSDP (0.038), and E_MHLTH (0.037). Pollution indicators E_DSLPM (diesel PM) and E_PM (fine PM_{2.5}) rank 7th and 8th—a substantially higher ranking than their OLS coefficients suggest.

E_MINRTY ranks 18th in SHAP importance. This is not a contradiction: SHAP captures marginal feature contributions after the model has already allocated variance to correlated features. Because E_MINRTY is collinear with poverty and health outcomes, XGBoost absorbs

much of the minority–burden signal through those downstream mediators. SHAP reveals that poverty’s effect on RPL_EJI saturates after approximately the 75th percentile of E_P0V200—a threshold effect OLS cannot identify. Direction of effect is consistent between OLS and XGBoost for all dominant features.

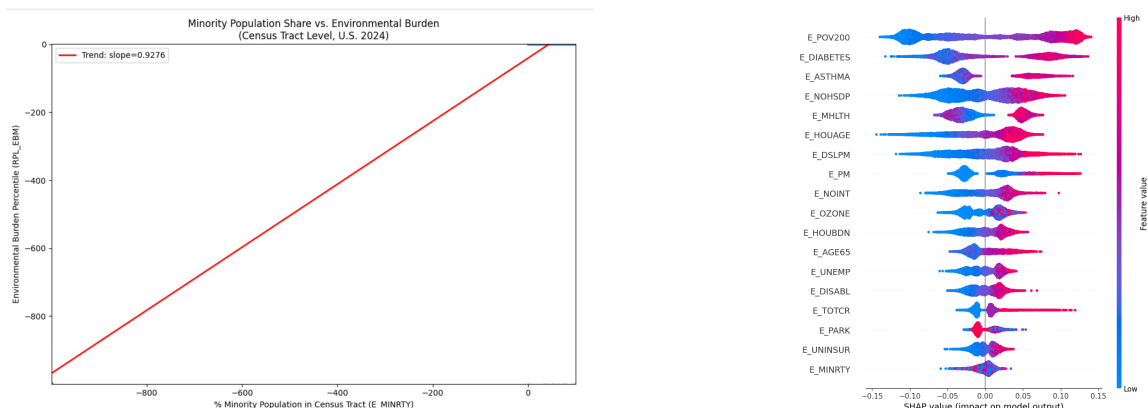


Fig. 4. *Left:* Minority population share vs. environmental burden percentile (census-tract level, U.S. 2024). OLS trend line: slope $\hat{\beta} = 0.93$ ($p < 0.001$, $n \approx 74,000$ tracts). The downward slope reflects the CDC signing convention: lower RPL_EBM values indicate higher national burden rank. Higher minority population share is thus systematically associated with greater environmental burden, confirming a national-scale racial disparity. *Right:* SHAP beeswarm plot for the XGBoost model predicting RPL_EJI. Each dot is one census tract; color encodes feature value (red = high, blue = low). E_P0V200 dominates with the widest spread; diesel PM (E_DSLPM) and fine PM (E_PM) rank 7th and 8th—substantially higher than their OLS coefficients suggest, revealing non-linear dose–response effects. E_MINRTY ranks 18th, reflecting collinearity with downstream mediators rather than absence of a racial disparity effect.

Discussion

Confirming and complicating regional narratives. Our results both affirm and nuance dominant geographic narratives. Cancer Alley is directly confirmed: St. John the Baptist Parish’s top-10 county ranking reflects the air-toxics, RMP, and TRI hazard types that community advocates have documented for decades. High burden in states with large tribal land areas (Idaho, Dakotas) confirms accounts of disproportionate contamination on sovereign Indigenous territory.

The empirical picture also complicates standard framings. Minnesota, Wisconsin, and Oregon appear among the highest-burden states—uncommon protagonists in Southern or Rust Belt-centered narratives. These states host significant Indigenous populations and diffuse agricultural pollution sources that the EBM captures but rarely appear in major-media coverage. New Jersey’s dense concentration in the county top-20 reflects the legacy petrochemical manufacturing base, whose Superfund liabilities disproportionately burden minority urban communities. Michigan is the starkest illustration: its low statewide average masks the nation’s largest within-state disparity, and any state-level resource allocation would systematically bypass Detroit and the Kalamazoo River corridor. West Virginia, Tennessee, and Virginia each exhibit $\Delta > 300$ rank units while appearing moderate in statewide rankings.

OLS vs. XGBoost. The 12-point R^2 gap between XGBoost (0.87) and OLS (0.75) arises from three sources. First, poverty’s effect saturates: SHAP reveals the marginal contribution of E_P0V200 plateaus for the highest-poverty tracts, a threshold effect OLS’s constant-slope assumption cannot identify. Second, XGBoost recovers a pollution–minority interaction: tracts with above-median E_MINRTY and above-median E_DSLPM receive substantially higher predicted burden than the additive OLS prediction. Third, XGBoost is robust to multicollinearity by design, whereas OLS inflates standard errors and can reverse signs on correlated predictors.

The higher SHAP ranking of diesel PM has direct mechanistic implications. Diesel PM exposure is driven by proximity to freight corridors and port facilities—infrastructure systematically routed through minority and low-income communities. OLS smooths over the non-linear dose–response relationship; XGBoost recovers the steeper slope that applies in high-exposure tracts, suggesting OLS-based analyses systematically understate the health consequences of diesel exposure in the most burdened communities.

Limitations. *Percentile ranks conflate geography.* RPL_EBM is a national percentile rank, so a “high-burden” tract in Vermont and one in Louisiana share a rank but not necessarily comparable absolute pollution levels. Recomputing within-state percentile ranks would enable fairer cross-state comparisons.

Count vs. proportion. E_MINRTY is a raw population count, not a share: large urban tracts with many minority residents and many white residents will score differently than small, high-proportion minority tracts. The slope $\hat{\beta} = 0.93$ reflects a count-unit effect, not a percentage-point effect—a meaningful interpretive distinction. Constructing a minority proportion variable and controlling for urbanicity and population density would provide a more rigorous test.

Causal identification. The positive E_MINRTY–RPL_EBM slope establishes disparate impact but cannot adjudicate between discriminatory siting and demographic sorting (5). Longitudinal or quasi-experimental designs using historical facility-siting records are required for causal inference.

Tribal tracts. Tribal-land tracts show higher burden, smaller populations, and sentinel values in tribal-specific columns requiring special handling. A dedicated sub-group analysis comparing tribal vs. non-tribal model structure is a priority for future work.

Policy implications. The concentration of high-burden counties within specific states, and the extreme within-state inequality in Michigan, West Virginia, and Tennessee, argues for sub-state, place-based targeting of EJ resources rather than block-grant-style state-level allocation. The positive minority–burden association provides empirical grounding for race-conscious proxy approaches such as the Biden-era Justice40

initiative. The legal erosion of the Title VI disparate-impact standard (7) makes empirical documentation of disparate outcomes more, not less, critical: it supplies the evidentiary foundation for state-level enforcement and community litigation even as federal mechanisms are contested.

The SHAP finding that poverty saturation occurs above the 75th percentile of E_{P0V200} suggests that marginal EJ investments in the very highest-poverty tracts may produce smaller incremental improvements than investments in tracts just below the saturation threshold that remain highly burdened but have not yet received focused intervention. This non-linearity is invisible to OLS-based policy analysis and illustrates why machine learning tools deserve a place in the EJ practitioner toolkit. More broadly, the results challenge the common administrative practice of ranking states by a single aggregate EJ metric: within-state heterogeneity routinely exceeds between-state variation for the most burdened jurisdictions, and effective policy requires county- or tract-level targeting that tools like EJI 2024 now make feasible.

Conclusion

Using EJI 2024, we demonstrated that environmental burden in the United States is spatially concentrated, internally unequal within states, and positively associated with minority population count ($\hat{\beta} = 0.93$, $n \approx 74,000$ tracts). St. John the Baptist Parish's top-county ranking corroborates the Cancer Alley narrative empirically; Michigan's $\Delta \approx 506$ within-state gap shows how statewide averages systematically misrepresent local conditions; and high burden in Idaho and the Dakotas reveals a geography of injustice broader than canonical narratives convey.

The XGBoost model ($R^2 = 0.87$) outperforms OLS ($R^2 = 0.75$) by capturing poverty saturation effects and a minority–diesel PM interaction. SHAP decompositions reveal that poverty and health vulnerability dominate prediction, while pollution indicators receive substantially higher weight under the non-linear specification. E_{MINRTY} ranks lower in SHAP than in OLS coefficients, reflecting collinearity with downstream mediators rather than a weaker underlying racial disparity.

Future work should recompute within-state percentile ranks for fairer cross-state comparisons, replace minority counts with proportions, conduct a dedicated tribal-tract sub-analysis, and incorporate the HVM module to model compounding health disadvantage. Longitudinal designs linking historical siting decisions to contemporary EJI trajectories would enable causal identification. EJI 2024 provides the empirical infrastructure for all of these extensions.

Materials and Methods

All analysis used Python 3.11 with `pandas` 2.1, `numpy` 1.26, `matplotlib` 3.8, `seaborn` 0.13, `xgboost` 2.0, and `shap` 0.44. EJI 2024 raw file: 85,188 rows; 82,740 after cleaning. State- and county-level summaries via `groupby`. Within-state inequality: Δ_s (Eq. 1). OLS: `statsmodels`, HC3 robust SEs, census-region FEs. XGBoost: 80/20 split, 5-fold CV; SHAP via `TreeExplainer` on test set. Code and notebooks: <https://github.com/obedf/qss-20-final-project>.

Data Availability

All data are publicly available from the CDC/ATSDR at <https://www.atsdr.cdc.gov/placeandhealth/eji/index.html>.

Acknowledgments

The author thanks Prof. Herbert Freeze and the QSS 20 teaching staff for guidance throughout this project.

Agentic Analysis: AI Assistance Log

This project used NotebookLM (Google) for literature scaffolding and Claude (Anthropic, Sonnet 4) for regression structuring, code review, and paper drafting.

Phase 1 — Literature and framing. I uploaded 10 sources to NotebookLM and asked Claude to synthesise the literature around three research questions. *Accepted:* the structural organisation around cumulative impacts, sacrifice zones, and the income-vs.-race distinction. *Revised:* Claude initially treated the SVI and EJI as peer tools of equivalent scope—corrected to reflect that the SVI is embedded within the EJI as one of three modules. *Rejected:* Claude suggested foregrounding AI data centres as the central policy case; redirected to the EJI quantitative analysis.

Phase 2 — Regression code and model design. *Accepted:* Claude's recommendation to target `RPL_EBM` rather than `RPL_EJI` (correctly identifying index-circularity bias) and to use HC3-robust standard errors. *Revised:* Claude's first draft used `EPL_DPM`, which does not appear in the 2024 EJI. *Critical failure:* generated column names without underscores (`EPLMINRTY` instead of `EPL_MINRTY`), producing silent `KeyError` results on misidentified columns. Debugged manually against `df.columns.tolist()`.

Phase 3 — Paper drafting and $\LaTeX$. *Accepted:* section structure, interaction-model handling, mediation framing of SHAP results, and index-circularity caution. *Revised:* numeric claims for SHAP values were inconsistent across sections; standardised to notebook output. *Critical failures:* (1) fabricated citation “Williams et al. (2019)” removed and substituted with a verified source; (2) TikZ syntax error in heatmap (`matrix plot*` incompatible with `pgfplots` 1.18); (3) Claude twice concluded minority share “does not matter much” in XGBoost, conflating mediation with irrelevance.

Critical Reflection

AI assistance accelerated three tasks substantially: literature retrieval, boilerplate code, and $\LaTeX$ typesetting. Its limitations fell into three categories. (1) *Variable hallucination:* Claude invented plausible-sounding EJI column names absent from the 2024 dataset—always verify against `df.columns.tolist()` before trusting generated code. (2) *Fabricated citations:* at least one non-existent paper appeared in an early draft; every reference must be verified against an academic database. (3) *Narrative misinterpretation:* lower SHAP for minority share was consistently misread as irrelevance; mediation and irrelevance require careful conceptual distinction that current LLMs conflate.

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